CIVIL INSPECTION REPORT

Site: Greenfield Residential Complex, Block C — Pune, Maharashtra
Report No: BSE-INS-2024-0471 | Date: 12 April 2025

1. Project Information

Client Name	Mr. Rajesh Mehta / Greenfield Developers Pvt. Ltd.
Site Address	Plot No. 47, Baner Road, Pune – 411045, Maharashtra
Structure Type	G+4 Residential Apartment Building
Year of Construction	2016 (Approx. 9 years old)
Inspection Date	10 April 2025
Inspection Agency	BuildSafe Engineering Pvt. Ltd., Pune
Lead Inspector	Er. Sunita Rao, M.E. (Structural), CEng
Report Reference	BSE-INS-2024-0471

2. Scope of Inspection

This inspection was commissioned to assess the structural integrity and overall civil condition of Block C following resident complaints about seepage, cracking, and visible surface deterioration. The inspection covered all common areas, external facades, terrace slab, staircase, parking podium, and selected internal units (Flats 101, 201, 301, 401).

Inspection methodology included visual examination, hammer-sounding (delamination detection), carbonation depth testing, cover-meter survey, and photographic documentation. Core samples were NOT extracted during this visit.

3. Area-wise Observations

3.1 Terrace Slab (Roof)

Condition	Poor
Cracking Pattern	Map/alligator cracking across 60–70% of terrace surface; crack widths ranging 0.3 mm to 1.8 mm.
Seepage	Active seepage observed at 4 locations penetrating through slab soffit into 4th floor ceiling. Dark efflorescence stains visible.
Waterproofing	Original bituminous membrane completely deteriorated; no functional waterproofing layer present.
Parapet Wall	Vertical cracks at 3 corners; spalling of plaster at base; reinforcement corrosion suspected.

Drainage	2 of 4 rainwater outlets partially blocked; ponding likely during monsoon.

3.2 External Facade — East & West Elevations

Plaster Condition	Bulging and hollow plaster in patches on 2nd and 3rd floor levels; total affected area approx. 18 sqm.
Paint / Coating	Chalking and peeling of exterior paint; UV degradation evident.
Window Sills	Drip grooves missing on most window sills causing water tracking along walls.
Expansion Joints	Sealant in horizontal expansion joint at 2nd floor lintel level has failed; gap exposed.

3.3 Staircase & Common Lobby

Staircase Soffit	Minor cracking (hairline) in stair soffit between G and 1st floor; no active leakage.
Lobby Floor	Tile debonding observed in ground floor lobby; hollow sound on tapping 4 tiles.
Handrail Fixing	One handrail bracket loose at 3rd floor landing; corrosion at base plate.

3.4 Parking Podium (Ground Level)

Slab Soffit	Corrosion stains and rust streaks on 6 beams; 2 beams show concrete delamination exposing rebar.
Column Base	Dampness at 3 column bases; possible upward moisture migration from ground.
Overhead Plumbing	Condensation on HVAC pipes causing dripping; waterproofing of pipe insulation required.

3.5 Internal Units — Flat 201 (Sample)

Bedroom 1 Ceiling	Brown staining and efflorescence from terrace seepage; plaster soft in 0.4 sqm patch.
Bathroom Wall	Wet patch behind WC wall panel; potential sanitary plumbing leak or seepage from adjacent flat.
Kitchen	No structural issues observed. Minor hairline cracks at ceiling-wall junction (shrinkage, non-structural).

4. Material & Structural Condition Assessment

Parameter	Observation	Standard / Limit
Carbonation Depth (avg)	22 mm (terrace beams)	< Cover depth (~25 mm)
Concrete Cover (min)	18 mm (parking beams)	>= 40 mm for beams
Rebound Hammer (Schmidt)	28–34 N/mm² equivalent	Design: M20 (20 N/mm²)
Crack Width (max)	1.8 mm (terrace slab)	Permissible: 0.3 mm
Rebar Corrosion (visual)	Active — parking beams	None acceptable
Waterproofing Integrity	Fully failed — terrace	Functional membrane required

5. Severity Assessment

Terrace Slab	**High**	*Active seepage, large cracks, absent waterproofing — risk of structural degradation if untreated.*
Parking Beams	**High**	*Exposed rebar with active corrosion; load-bearing capacity potentially compromised.*
External Facade	**Medium**	*Loose plaster poses falling risk; moisture ingress accelerating.*
Staircase Soffit	**Low**	*Hairline cracks only; monitor but no immediate repair needed.*
Lobby Tiles	**Low**	*Aesthetic and trip-hazard concern; no structural implication.*

6. Recommended Actions

Immediate (0–3 months):

1. Commission a detailed structural engineering assessment of the parking podium beams where rebar corrosion is active. Provide temporary shoring if required pending repair.

2. Remove failed waterproofing membrane from terrace and apply a fresh APP/SBS modified bituminous membrane system with minimum 2-layer application and fibre mesh reinforcement.

3. Seal all cracks wider than 0.3 mm using epoxy injection grouting. Apply crystalline waterproofing slurry on terrace slab after crack treatment.

Short-term (3–6 months):

4. Hack off all hollow and bulging plaster on external facades. Re-plaster with polymer-modified cement mortar and apply elastomeric exterior coating.

5. Replace failed sealant in expansion joints using polyurethane sealant with backer rod.

6. Clear blocked rainwater outlets and extend downpipes to prevent future ponding.

Long-term (6–18 months):

7. Carry out full Ferro-cement or jacketing repair of corrosion-affected beams after chloride content testing and carbonation profiling.

8. Review and upgrade column base waterproofing in parking. Install sub-surface drainage channel if water table is elevated.

7. Disclaimer & Inspector Declaration

This report is based on visual and non-destructive inspection carried out on the date(s) mentioned. It does not constitute a full structural design assessment. Findings are limited to areas accessible at the time of inspection. BuildSafe Engineering Pvt. Ltd. accepts no liability for hidden defects not apparent during this inspection.

Prepared by: Er. Sunita Rao, M.E. (Structural)

Reviewed by: Er. Anil Deshmukh, Chief Engineer — BuildSafe Engineering Pvt. Ltd.

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